Submodule 33.06: Feature Correlation Matrix & Heatmap Analysis (Hinglish)

Topic Overview & Mathematical Formulation

Encoding aur scaling se pehle features ke aapas ke linear relationships ko check karna zaroori hota hai. Iske liye hum **Pearson Correlation Coefficient (r)** calculate karte hain:

$$r_{XY} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}}$$

- $r = +1$: Strong positive correlation (ek feature badhega toh dusra bhi badhega).
- $r = 0$: No correlation (dono features aapas mein independent hain).
- $r = -1$: Strong negative correlation (ek feature badhega toh dusra ghatega).

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1. Python Code Implementation

Reference: smartcart.ipynb (Cells 24 - 28).

```
import matplotlib.pyplot as plt
import seaborn as sns

# Numeric features ka correlation matrix calculate karna
corr = df_cleaned.corr(numeric_only=True)

# Heatmap plot karna
plt.figure(figsize=(8, 6))
sns.heatmap(
    corr,
    annot=True,
    annot_kws={"size": 6},
    cmap="coolwarm"
)
plt.title("Correlation Heatmap")
plt.show()
```

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2. Heatmap se Mili Key Business Insights

Feature Pair	Correlation (r)	Nature	Business Explanation
`Income` vs `Total_Spending`	**+0.79**	Strong Positive	Dataset ka sabse strong correlation! Income badhegi toh total spending bhi proportionally badhegi.
`Total_Spending` vs `NumCatalogPurchases`	**+0.78**	Strong Positive	Jo log zyada spend karte hain, woh catalog se direct khareedte hain.
`Income` vs `NumCatalogPurchases`	**+0.69**	Positive	High income wale customers catalogs se purchase karte hain.
`Income` vs `NumStorePurchases`	**+0.63**	Positive	High income customers physical stores mein visit karke shopping karte hain.
`Income` vs `NumWebVisitsMonth`	**-0.65**	Strong Negative	High income customers website par bar-bar browse nahi karte; low-income log website bar-bar browse karte hain (window shopping).
`NumStorePurchases` vs `NumWebVisitsMonth`	**-0.43**	Moderate Negative	Jo log physical store se shopping karte hain, unki monthly website visits kaafi kam hoti hain.
`Customer_Tenure_Days`, `Complain`, `Response`	≈ 0.0	Independent	Yeh features independent information provide karte hain (white color in heatmap).

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3. Takeaway

Correlation matrix ne prove kiya ki dataset mein do prominent dimensions hain:

1. **Financial Power**: Income, Total Spending, Catalog Purchases.
2. **Channel Behavior**: Store & Catalog vs Repeated Website Browsing. Yeh collinearity confirm karti hai ki **PCA (Principal Component Analysis)** dimensionality reduce karne ke liye perfectly suited hai!